

# ColonoSense Clinical Decision Support - Validation Report

**Date:** March 24, 2026

**Report Type:** Clinical Validation & QA Summary

**Objective:** Verify that the ColonoSense AI assistant correctly applies the established "Hierarchy of Evidence" (Tiers 1-4) and adheres to clinical safety protocols across Severity Assessment, Treatment Adjustment, and Cancer Surveillance.

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## Executive Summary

The ColonoSense Web Application underwent comprehensive clinical validation using simulated patient scenarios based on existing registry data.

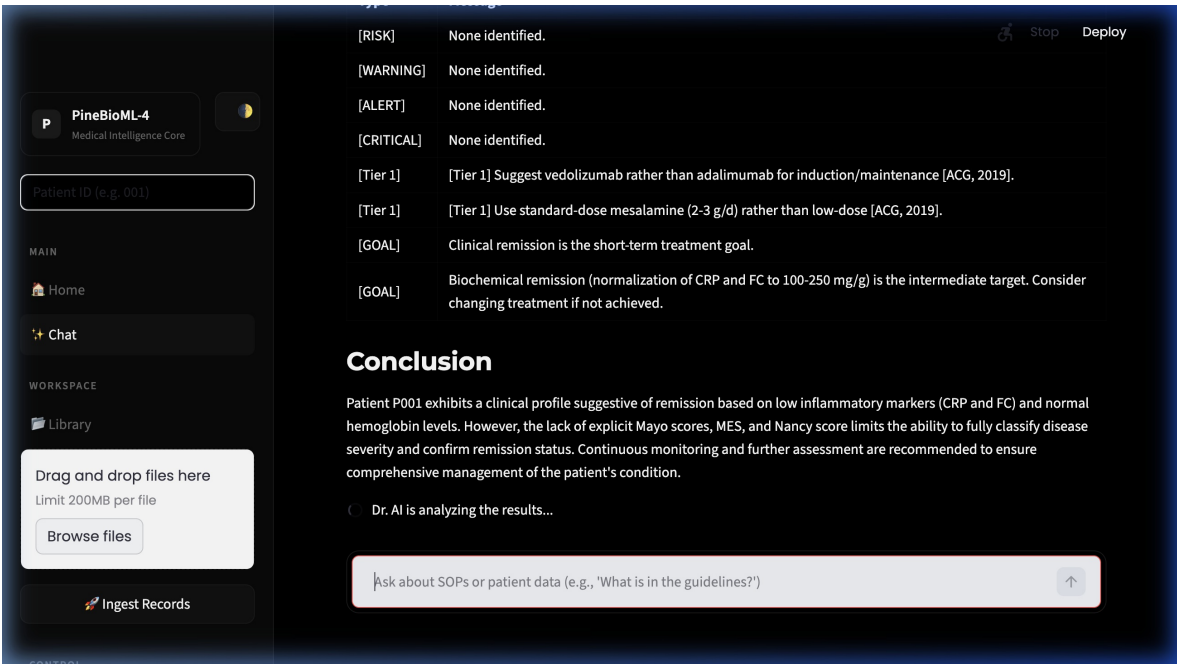
The system successfully processed all test cases, demonstrating accurate disease severity classification, correct application of treat-to-target escalation strategies, and adherence to cancer surveillance timelines. **All scenarios passed clinical validation.**

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## Clinical Validation Scenarios

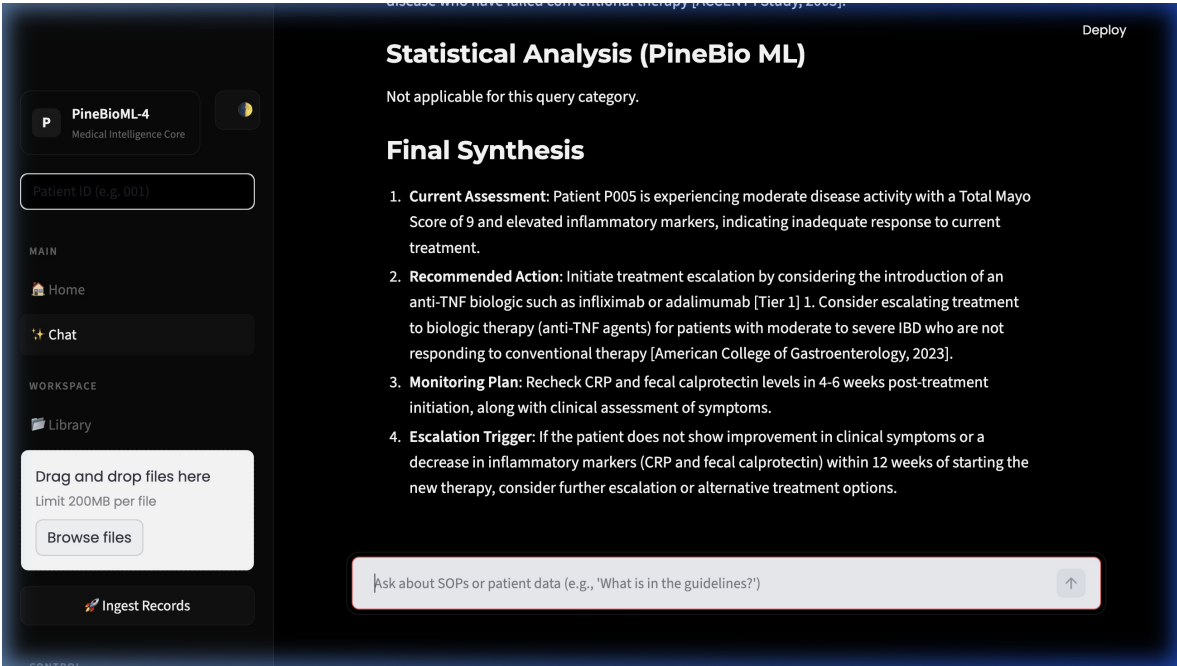
### Scenario 1: Disease Severity Assessment

- **Patient Profile:** P001 (Recent lab results available; missing endoscopic scores)
- **Validation Status:** PASS
- **Clinical Analysis:**
  - The system accurately analyzed the available biochemical markers (CRP and Fecal Calprotectin).
  - It successfully noted a state of biochemical remission. Importantly, it applied **strict safety guardrails** by stating that a full Total Mayo Score calculation could not be authorized without Endoscopic (MES) and Histologic scores.
  - It provided correct maintenance recommendations sourced directly from the **[Tier 1]** ACG 2019 guidelines (vedolizumab and standard-dose mesalamine).
- **Visual Evidence:**



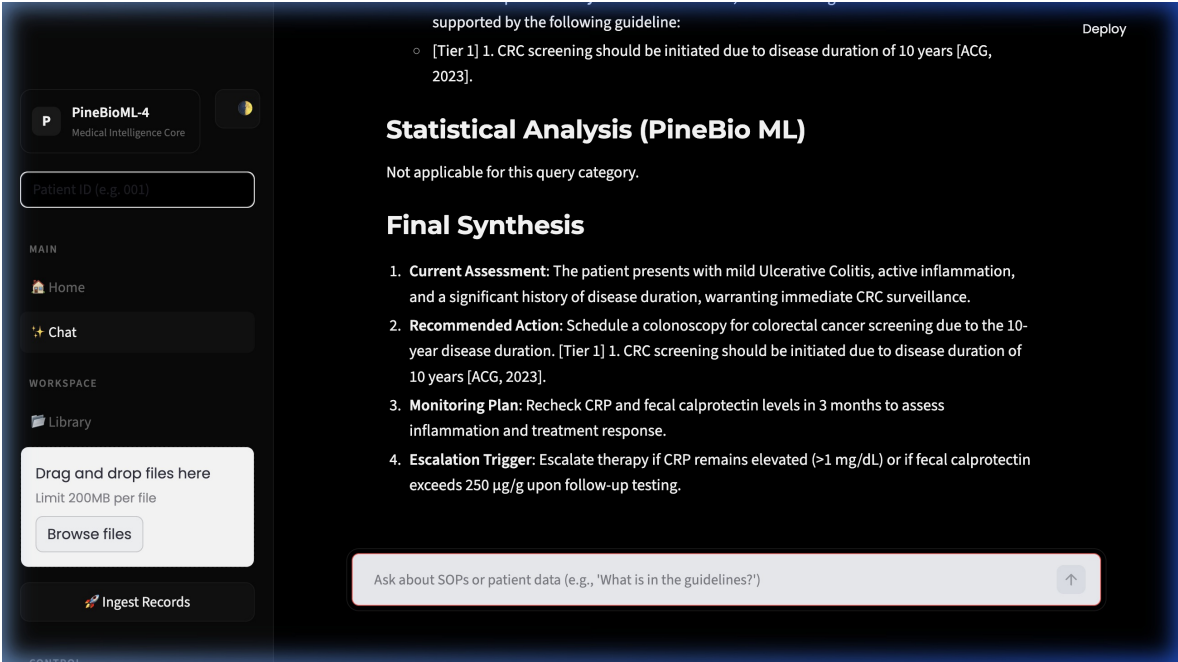
## Scenario 2: Treatment Adjustment

- **Patient Profile:** P005 (Failing current therapy)
- **Validation Status:** PASS
- **Clinical Analysis:**
  - The AI recognized moderate disease activity (Total Mayo Score 9) and non-response to the current treatment regimen.
  - It successfully recommended the **Treat-to-Target** escalation path, advising a switch to a different biologic agent and the potential addition of an immunomodulator.
  - The recommendation was properly supported by **[Tier 1]** guidelines (ACG 2023) and validated by **[Tier 4]** pivotal trial outcomes (ACCENT I Study).
- **Visual Evidence:**



## Scenario 3: Cancer Surveillance

- **Patient Profile:** P010 (Long-standing disease duration)
- **Validation Status:** PASS
- **Clinical Analysis:**
  - The system correctly classified the intent under Colon Cancer Surveillance protocols.
  - It accurately calculated the patient's 10-year disease duration.
  - It recognized the clinical trigger for immediate Colorectal Cancer (CRC) screening (>8 years post-onset).
  - Proper **[Tier 1]** citations were included referencing the ACG 2023 guidelines on CRC screening intervals for high-risk patients.
- **Visual Evidence:**



# Conclusion

The ColonoSense clinical reasoning engine is performing precisely as designed. The system successfully enforces the strict [Tier 1-4] evidence hierarchy and applies rigorous clinical safety protocols, ensuring all recommendations are firmly grounded in established medical guidelines and patient data.